In-class Assignment

IC: Case 31 - Summary of Data Analysis: R1, R2, R5, & R6

Using R, obtain appropriate descriptive and inferential statistics. Using QMD file, summarize the data analysis for R1, R2, R5, and R6 for Case 31. In doing so, insert appropriate R output into the HTM document. Refer to the output while you describe the findings.

Caution: The most important thing in data analysis is to generate insight. Don't let statistics get in the way. For every research question or hypothesis, look at descriptive statistics first. That is, look at frequencies or means, percentile, median, or skewness, first. Next, think about what story the data tell you. After this, you may conduct inferential statistics to see if the statistics confirm what you think. When you present to your clients, you don't even talk about inferential statistics. Most of the time, you will deal with frequency and percentage associated with it.

Tasks:

- Save QMD file with your name part of it
- Organize your QMD report file using appropriate research question numbers (e.g., R1, R2, etc.) and type of test you are doing (e.g., Descriptive Statistics, one sample t-test, Ch-square goodness-of-fit test, etc.).
- Per each research question, state the insight you generated. Your response should address the research question. Your response should not be just a five-step sequence of hypothesis test. You may report test statistics (e.g., t-value, chi-square value) and p-values associated with it, but should do so inside parenthesis so that statistics do not get in the way when readers read it. This is how you would present orally too. You do not want to talk about hypothesis test orally. Just showing it on the slide is enough.
- Refer to the output while you describe the findings. To better communicate in written report, you are encouraged to use table or figure numbering in Quarto using "tbl-xxxxx" or "fig-xxxxxx" inside a code chunks with a hash pipe.

Sample of Results (Ch4) Section

Below is an excerpt from one of my published paper (Jung et al., 2020) in which I used cross-tabulation and chi-square independence test. Take a look at highlighted area to understand my point regarding how statistics should not get in the way when you tell a story.

Analysis and Results

Since both independent and dependent variables are frequencies of dichotomous nominal variables, Chi-square tests of association were used to test the hypotheses. Thus, a series of 2 (dependent variable) x 2 (SOO) x 2 (residency) cross-tabulation was performed. Also a series of 2 (residency) x 2 (SOO) x 2 (dependent variable) cross-tabulation was performed. Table 2 shows variables, their variable levels, their frequencies, chi-square statistics within each state, and chi-square statistics across the states within each variable level.

Table 2
Chi-Square Test of Association:
2 (Dependent Variable) x 2 (SOO) x 2 (Residency) Cross-Tabulation

		North Dakota residents only (N = 121)			*Ohio residents only (N = 126)*			*Across the State*	
		State of Origin			*State of Origin*				
Dependent Variables	*Levels*	Pride of Dakota	Ohio Pride	χ^2 value[a]	Pride of Dakota	Ohio Pride	χ^2 value[a]	χ^2 value[b]	
Attitude towards products (H2)	Favorable	54	40	8.78 (p=.012)	38	54	14.49 (p=.001)	4.85	*
	Unfavorable	2	9		17	4		11.79	***
	Neutral	5	11		10	3		5.99	*
Preference (H3)	Yes	29	10	12.94 (p=.002)	7	24	13.92 (p=.001)	18.54	***
	No	18	32		36	20		8.46	**
	Neutral	16	15		19	18		.00	
	Others[c]	0	4		3	0		--	
Obligation/ Loyalty (H4)	Yes	23	5	15.05 (p<.001)	1	29	36.26 (p<.001)	37.08	***
	No	33	50		56	27		12.81	***
	Neutral[c]	5	1		0	4		--	
	Others[c]	2	4		8	2		--	
Attitude - Positive factors (H5)	Product quality (1)	7	7	--	12	10	--	.07	
	Logo design (2)	6	24		29	9		21.29	***
	Feelings about state (3)	2	6		2	1		--	
	SOO (4)	27	5		5	17		20.52	***
	Support local economy (5)	11	3		1	14		15.44	***
	COO (6)	1	8		6	1		--	
	Others (7)	6	6		8	8		.00	
	Cognitive (1+2)	13	31	16.77 (p<.001)	41	19	17.13 (p<.001)	18.61	***
	Affective (3)[c]	2	6		2	1		--	
	Normative (4+5+6)	39	16		12	32		30.11	***

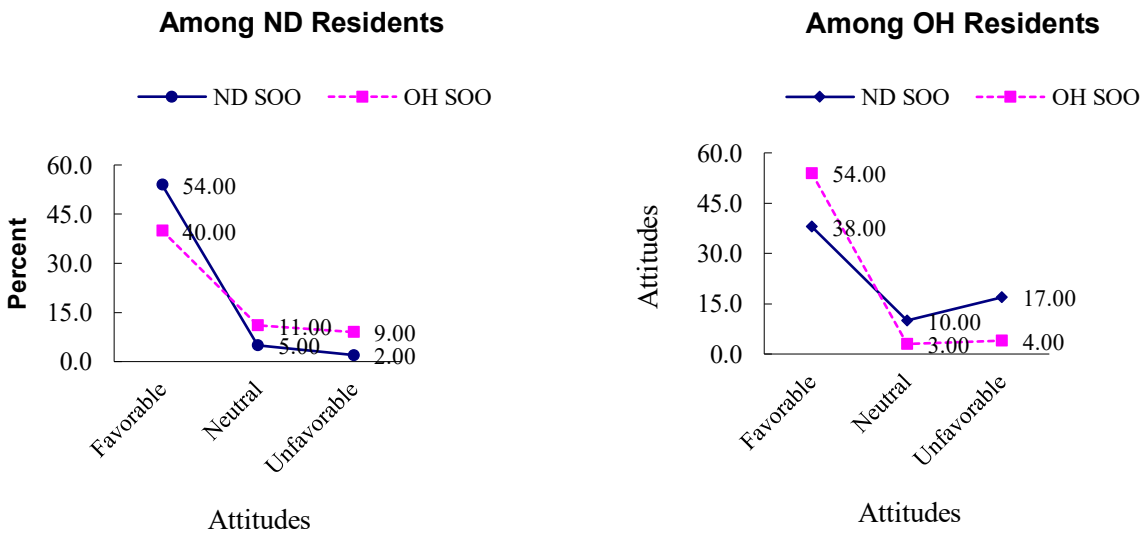
Note: * $p < .05$; ** $p < .01$, *** $p < .001$. a: χ^2 value within the resident groups. b: χ^2 value across residents groups for each dependent variable level and based on 2 x 2 table ($df = 1$). c: indicates a variable level that is dropped due to small expected cell numbers (n < 5), in which case, χ^2 value[a] was calculated without the dropped level(s) and χ^2 value[b] is not reported.

Attitude Toward Products (H2)

Results of Chi-square test show that SOO influences attitude toward products bearing state logo among both ND residents ($\chi^2_{(2)} = 8.78$, $p = .012$) and OH residents ($\chi^2_{(1)} = 14.49$, $p = .001$). In addition, the frequency pattern of attitude for products bearing ND vs. OH logo is the opposite across the two state resident groups significantly for each level of the three attitude category (Favorable: $\chi^2_{(1)} = 4.85$ $p < .02$; Unfavorable: $\chi^2_{(1)} = 11.79$ $p < .001$; Neutral: $\chi^2_{(1)} = 5.99$, $p < .01$) as seen Table 2 and Figure 2. Thus, a series of the patterns of results showed that residency moderated SOO effects such that participants favored products bearing their own state's (vs. other state's) logo, which is consistent with results of study 1, supporting H2.

FIGURE 2

Attitude towards products bearing state logos



Preference (H3)

Results of Chi-square test show that SOO influences preference for products bearing state logos among both ND residents ($\chi^2_{(2)} = 12.94$, $p = .002$) and OH residents ($\chi^2_{(2)} = 13.92$, $p = .001$). Among ND residents, 46.0% of those presented with their own state logo expressed their preference for the products bearing the logo, while 17.5% of those presented with the OH logo expressed their preference for the products bearing the logo. Likewise, among OH residents, 38.7% of those presented with their own state logo expressed their preference for the products bearing the logo, while 11.3% of those presented with the ND state logo expressed their preference for the products bearing the logo. Moreover, the frequency pattern of the participant's *preference* for products bearing ND vs. OH logo is reversed across the two resident groups on "preference" ($\chi^2_{(1)} = 18.54$, $p = .000$) and "no preference" ($\chi^2_{(1)} = 8.46$, $p < .01$) levels but not on "neutral" levels ($\chi^2_{(1)} = .00$, *NS)* as shown in Table 2. Thus, a series of the patterns of the frequencies within and between the two resident groups showed that residency moderated the SOO effect such that participants have clear preference for products bearing their own state over other state, consistent with H3.